

JOSEPH ALAN B. VERGARA

• Phase II-Doña Maria Subd, Tubod, Lanao Del Norte • <https://www.joalvergs.tech/> • (+63) 947-589-2995

EDUCATION

BS Computer Applications Major in Embedded Systems

4th Year Student at MSU- Iligan Institute of Technology | **CGPA: 1.96269**

August 2023 - Present

Iligan City, Philippines

Senior High School | ABM Strand

2021-2023

MSU-IIT Integrated Developmental School | **Academic Honors**

A. Bonifacio Ave, Brgy. Tibanga, Lanao Del Norte

SKILLS

Programming Language:	C, C++, Python, JavaScript, TypeScript, Java, PLC (Ladder Logic), Assembly, Bash/Shell Scripting, CSS
Operating Systems & CLI:	Linux (Arch Linux, CachyOS), Unix Shell & CLI Configuration (Zsh, Bash, Dotfiles), RT-Thread RTOS
Networking & Web Infrastructure:	DNS Management (A, CNAME, TXT records, Nameservers), Custom Domain Configuration, SSL/TLS, Vercel, Railway, Hostinger, Cisco Packet Tracer
Software & Frameworks:	Visual Studio Code, Git, GitHub, Node.js, React.js, Express.js, Pictoblox, CodeBlocks, Microchip (ATMEL) Studio, Arduino IDE, Google Colab, Edge Impulse, TensorFlow, MATLAB, MARIE.js, Vivado, RT-Thread Studio, SquareLine Studio, MongoDB, Supabase
Hardware:	Microcontrollers (i.e., 8051, ATmega328P, ATmega2560, ESP8266, ESP32), Field Programmable Gate Arrays (FPGAs), Basic and Advanced Electronics, Test Equipment (i.e., Multimeter, Oscilloscope), Prototyping (i.e., PCB layout, design, prototyping, simulation, soldering), Computer & Laptop Hardware Repair (Teardown, Component Diagnostics, Parts Replacement)
Electronics:	Applied Math & Calculus, Principles of Electric Circuits, Electronics Fundamentals, Digital Design, Computer Architecture
Soft Skills:	Cross-Functional Team Collaboration Abilities, Organizational & Time Management Skills, Being Self-Driven and Independent, Adaptability & Patience
Spoken Language:	English, Filipino, Cebuano

PROJECTS

Microcontroller-Based Automated Feedback and Scoring System (In Progress)

Co-Author & Full-Stack / Embedded Developer | PhD Dissertation Research Collaboration

- Designed a specialized full-stack Learning Management System (React.js, Node.js, Supabase/PostgreSQL) to capture live microcontroller telemetry and automate laboratory scoring logic.
- Reduced evaluation latency and eliminated manual record-keeping errors by engineering deterministic evaluation pipelines paired with interactive instructor dashboards for instant feedback.
- Benchmarked standard web server deployments against an edge-hosted architecture on the ESP32-P4 microcontroller to evaluate low-power, localized server reliability. | *Live Demo:* www.miow-lms.ids-ter.online | *GitHub:* <https://github.com/MIOW-CODES/lms>

Mobile Application Development for Computer Vision AI Development in Detecting Microplastics (In Progress)

Co-Author & Mobile/Edge AI Developer | Master's & Undergraduate Research Collaboration

- Deployed an optimized YOLOv8n edge AI model into a cross-platform mobile app for zero-latency, on-device particle quantification.
- Configured Android SDK build environments to generate release APKs and established deployment bundles for iOS targets.
- Enhanced UI/UX flow to handle camera lifecycles, live bounding-box rendering, and research-grade inference visualization. | *Documentation:* https://drive.google.com/drive/folders/1GOxtqUJavVIL_n6gHiRzwwTUOWCLO-tB?usp=sharing | *GitHub:* <https://github.com/MP-DETECT-CODE>

Barangay Connect Web Application

Full-Stack Developer | Web Development, GitHub

- Built a full-stack portal streamlining local public service requests, document generation, and real-time community reporting. | GitHub: <https://github.com/Joal0816/Barangay-Connect>

Edge AI Glasses Object Detection

Edge AI Developer | Roboflow, Edge Impulse

- Trained, quantized, and validated a low-latency edge vision model for wearable hardware via Edge Impulse and custom Roboflow datasets. | Project: <https://studio.edgeimpulse.com/studio/848525>

Industrial Oven Simulation & Control System

PLC & Automation Developer | CODESYS V3, PLC, HMI

- Engineered virtual PLC ladder logic and an interactive HMI with automated heating logic, safety interlocks, and threshold alarms. | Documentation: <https://sites.google.com/g.msuiit.edu.ph/hassanvergarafinalproject?usp=sharing>

Snake OS: Embedded Interactive HMI Game

Embedded Developer | Renesas RA6M3, RT-Thread OS, LVGL

- Engineered an arcade game engine utilizing RT-Thread RTOS and LVGL to deliver fluid real-time graphical rendering on bare metal.
- Implemented autonomous AI autopilot navigation, non-volatile high-score persistence, and dynamic multi-level obstacle logic. | Documentation: <https://www.hackster.io/josephalanvergara/snake-os-interactive-hmi-game-on-rt-thread-f8b988>

IoT-Based Monitoring System for Vermicomposting

IoT Developer | ESP32, MQTT, Sensors

- Automated closed-loop irrigation based on real-time soil moisture (60–80%) and piped 5+ environmental parameters over MQTT. | Documentation: <https://canva.link/k5sngerg43lbggp>

Line Following Robot

Firmware Developer | Arduino Uno, AVR Assembly

- Hand-tuned AVR Assembly firmware and dual PWM hardware timers (Timer0/2) for a 5-channel IR array, achieving a 100% course completion rate. | Documentation: https://docs.google.com/document/d/1k9SudemZJafofm2p3hludQYK7nMUtvUpkMm6N-_rJvY/edit?usp=sharing

K-Bin: Automated Smart Waste Sorting System

Software & Embedded Developer | Qt C++, CMake, Sensors

- Developed a Qt C++ GUI with real-time level tracking and multi-sensor routing (inductive, capacitive, ultrasonic) to sort 3 waste streams. | Project Files: https://drive.google.com/drive/folders/1OVPIXr5QFTa3_Uin5QVysLJvsKLshUdQ?usp=sharing

Adjustable DC Power Supply: Engineered a regulated benchtop power supply unit with variable output up to 12V and short-circuit protection.

TRAININGS & WORKSHOPS (CERTIFICATIONS)

AI Career Readiness Training

August 20, 2026

ICT 3D, CCS Building, MSU-IIT, Iligan City

Quantum Circuits as Pictures: An Introduction to ZX Calculus

March 20, 2026

Zoom Meeting, Webinar

Permaculture Webinar “From Code to Crops”

December 22, 2025

Zoom Meeting, Webinar

TechShowcase 2025 Innovation in Application Development and Emerging Technologies

December 17, 2025

PRISM Minitheater, MSU-IIT, Iligan City

my.ComApps Technology Symposium and Exhibit on RT-Thread Based Technology

November 24-25, 2025

4th floor, CCS Building, MSU-IIT, Iligan City

Seize Your Opportunity Internships and OJTs Workshop!

November 24, 2025

ICT 3D, CCS Building, MSU-IIT, Iligan City

Software Freedom Day 2023

September 16, 2023

CCS Building, MSU-IIT, Iligan City

LEADERSHIP & ACTIVITIES

College of Computer Studies Executive Council - Sports & Development Affairs - Undersecretary

August 2025 – May 2026

MSU- Iligan Institute of Technology, Iligan City, Philippines

Titans Esports - Mobile Legends BangBang Team Captain

2021-2023

MSU- Iligan Institute of Technology, Iligan City, Philippines